School of Computing & Engineering 5-Year Strategic Development Plan (2025-2030)

Office of the Dean & Provost | Research Excellence, Infrastructure & NIRF/ABET Roadmap

1. Executive Vision & Strategic Intent 2025-2030

The School of Computing and Engineering at Northgate Institute of Technology presents this Five-Year Strategic Development and Institutional Governance Plan to the Board of Governors for approval and funding commitment. The plan positions the School to achieve: (a) entry into the NIRF Top-50 Engineering Institutions ranking by 2027 (current position: Rank 87); (b) achievement of NIRF Top-30 by 2030; (c) completion of ABET Tier-1 accreditation for all B.Tech and M.Tech programmes by Winter 2026; (d) establishment as a nationally recognized center for applied AI research and responsible computing. The plan is aligned with NEP 2020 multidisciplinary learning mandates, the IndiaAI Mission's capacity-building objectives, and the 'Viksit Bharat @ 2047' technology readiness goals.

NIRF ranking improvement strategy draws on the methodology-aware approach recommended by education consulting firms: the School's primary focus areas are Teaching, Learning & Resources (TLR — 30% NIRF weight) and Research & Professional Practice (RP — 30% NIRF weight), which together constitute 60% of the composite score. Secondary targets are Graduation Outcomes (GO — 20%), Outreach & Inclusivity (OI — 15%), and Perception (PR — 5%). Data integrity for NIRF submissions is managed by a dedicated NIRF Data Management Cell established under the Dean's office.

2. Strategic Pillar 1: AI Supercomputing Infrastructure (Budget: Rs. 14.2 Crores)

The School has commissioned the establishment of the High-Performance AI Supercomputing Center (HPAISC) — 'Project Swarnim' — co-funded through a DST-FIST matching grant (Rs. 6 Crores) and University capital expenditure (Rs. 8.2 Crores). Phase 1 (Q1 2026): Procurement and installation of 8 interconnected NVIDIA H100 SXM5 GPU compute nodes (80GB HBM3 per node) delivering over 32 petaflops of FP8 tensor computing capacity, interconnected via 400 Gbps InfiniBand NDR fabric. Phase 2 (Q3 2026): Integration of a 4 PB all-flash NVMe storage cluster (Dell PowerScale F910) for high-throughput ML training datasets. Phase 3 (Q1 2027): Establishment of an AI Model Repository and collaboration portal enabling shared access for 6 partner universities under the National Supercomputing Mission (NSM) collaborative network.

The HPAISC will provide dedicated computational allocations for: doctoral dissertations (priority queue, 24h allocation per active Ph.D. scholar/week), faculty PI-approved research grants (proportional to grant quantum), industry-sponsored AI consortia (premium paid access, supporting the School's tech-transfer revenue model), and B.Tech Honors track students (2h/week allocation for capstone research). An Ethical AI Governance Committee will oversee access policies and ensure compliance with the IndiaAI Mission's Responsible AI framework.

3. Strategic Pillar 2: Faculty Recruitment & Endowed Chair Programme

To improve TLR scores and research capacity simultaneously, the School will execute the most ambitious faculty hiring programme in its history: 18 tenure-track Assistant Professor positions and 4 Distinguished Endowed Chair positions over 5 years (2025-2030). Target specializations: (a) Generative AI & Foundation Models (4 positions); (b) Quantum Computing Architectures (2 positions); (c) Hardware Security & Trusted Execution Environments (2 positions); (d) Edge Computing & IoT Systems (2 positions); (e) Cybersecurity & Zero-Trust Architecture (2 positions); (f) Data Engineering & LLMOps (2 positions); (g) Human-Computer Interaction & Assistive Technology (2 positions); (h) Bioinformatics & Computational Genomics (2 positions, multi-disciplinary with Life Sciences).

Endowed Chair holders (funded at Rs. 25 Lakhs/year research discretionary from the University Innovation Endowment) must: (a) lead one of the 3 Centers of Excellence; (b) attract minimum Rs. 1 Crore in external grants within 3 years; (c) mentor 2 early-career faculty members annually; (d) deliver 2 public distinguished lectures per year. International recruitment campaigns will target returning Indian diaspora researchers from leading institutions (MIT, Stanford, CMU, Google Research, Microsoft Research, DeepMind) through the 'Vasudhaiva Kutumbakam' Faculty Homecoming Programme.

4. Strategic Pillar 3: Three Centers of Excellence (CoEs)

- **Center for Trustworthy AI & Algorithmic Ethics (CTAAE):** Directed by Dr. Ananya Krishnan. Research focus: safety, fairness, robustness, and explainability of AI systems deployed in critical decision-making contexts (healthcare triage, credit scoring, criminal justice, university admissions). Founding industry partners: Persistent Systems, Wipro AI Labs. Target: 5 NeurIPS/ICML/FAccT papers/year by Year 3; Rs. 2 Crores DST-funded project by Year 2.

- **Center for Cyber Defense & Zero-Trust Architecture (CCDZT):** Directed by Dr. Julian Vance. Focus: critical infrastructure protection, hardware-rooted trust, automotive cybersecurity (ISO 21434), and quantum-safe cryptography transition. Partners: CERT-In, DRDO, Bosch Engineering. Target: quarterly CTF (Capture The Flag) competition for students and industry practitioners; 2

DRDO-funded projects by Year 2.

• **Center for Sustainable Cloud & Edge Systems (CSCES):** Directed by Dr. Ravi Subramaniam. Focus: carbon-aware distributed computing, energy-proportional data centers, workload scheduling for renewable energy integration, edge inference efficiency (quantization, pruning, neural architecture search). Partners: AWS Sustainability, Microsoft Azure Research, NVIDIA AI Enterprise. Target: 2 IEEE/ACM systems conference papers/year; ISO 50001 energy management certification for HPAISC facility by Year 2.

5. Strategic Pillar 4: ABET & NAAC Accreditation Roadmap

The School aims to complete ABET accreditation (Criteria for Programs in Computing — CAC) for all B.Tech and M.Tech programmes by Winter 2026. The Department Curriculum Committee has been restructured as a standing body meeting monthly, responsible for: (a) mapping all Course Outcomes (COs) to ABET Student Outcomes (a)–(j) and AICTE Graduate Attributes; (b) executing direct and indirect assessment of CO attainment each semester; (c) closing the loop through documented curriculum revisions based on assessment data. NAAC institutional accreditation renewal (target: Grade A++, current: A) requires submission of the Institutional Development Plan (IDP) by December 2025 and a site visit in Q2 2026. The School contributes data for 4 NAAC criteria: Curricular Aspects (C1), Teaching-Learning & Evaluation (C2), Research, Innovation & Extension (C3), and Infrastructure & Learning Resources (C4).

6. Strategic Pillar 5: Industry Consortia, Tech Transfer & Incubation

The School is establishing a formal Industry-Academia Technology Consortium (IATC) with 10 founding corporate members (Microsoft, AWS, Infosys, Qualcomm, Bosch, TCS, Wipro, HCL, ISRO-SAC, and DRDO-CAIR) each contributing Rs. 25 Lakhs/year in consortium fees for access to: dedicated research collaboration with faculty, priority campus recruitment access, co-development of 2 specialized elective courses annually, and IP licensing right-of-first-negotiation on School-developed patents.

The Student Innovation & Incubation Center (SIIC) expansion plan (Phase 2): increase active incubated startups from 12 to 30 by Year 3; establish an Alumni Angel Network (target: 100 accredited investors by Year 2); launch the Deep-Tech Accelerator Programme in partnership with NASSCOM 10,000 Startups (12-week cohort, 2 cohorts/year, Rs. 5 Lakhs seed + cloud credits + mentorship). The School's patent filing rate target: 10 provisional patents/year by Year 3.

7. Strategic Pillar 6: Diversity, Equity & International Partnerships

The School commits to increasing female student enrollment in engineering programmes from the current 28% to 40% by 2030 through: (a) the WTLA scholarship expansion (increasing awardees from 10 to 25); (b) Girls4Tech outreach programme visiting 100 partner schools annually in Tier-2 and Tier-3 cities; (c) women-only hackathon 'CodeHer' with Rs. 5 Lakhs prize pool; (d) mentorship matching with women alumni in senior industry positions. International partnerships: expand MoU exchange network from 14 to 25 partner universities by 2028, with particular focus on ASEAN, European, and African institutions to fulfill NEP 2020's internationalization mandate.

8. Research Grant & IDC Revenue Model

Indirect cost overheads (IDC) from external grants are distributed per the University Research Finance Policy: 50% to the Central University Research Facilitation Fund (CURFF) supporting shared infrastructure; 30% to the Host Department for equipment, lab consumables, and RA stipends; 20% to the PI's Professional Development Account for conference travel, equipment purchases, and researcher training. The School's aggregate externally funded research portfolio target: Rs. 15 Crores/year by Year 5 (current: Rs. 4.2 Crores/year). Faculty eligible for Research Sabbaticals (1 year, paid, extendable by 6 months unpaid) after 6 consecutive service years, contingent on completion of all current student supervision commitments.